Markov chain Monte Carlo, part 3

Covered by chapter 11 in the textbook and these notes.

The Gibbs Sampler Up to now, we have considered MCMC methods which update the complete state vector $\boldsymbol{\theta}$ in each iteration of the algorithm. This works well in low dimension (and in certain cases in high dimension as well). Now we are going to look at methods for "dividing" the original sampling problem into a sequence of smaller, and hopefully simpler problems: The Gibbs sampler.

Overarching idea: Let the state $\boldsymbol{\theta}$ be partitioned into two (or in general more) sub-vectors $\boldsymbol{\theta} = (\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})$. Sub-vectors referred to as "blocks".

Then alternate between updating the blocks according to the **conditional** distributions:

- Input: old state $\boldsymbol{\theta}_{t-1} = (\boldsymbol{\theta}_{t-1}^{(1)}, \boldsymbol{\theta}_{t-1}^{(2)})$
- Update $\boldsymbol{\theta}_t^{(1)} \sim \pi(\boldsymbol{\theta}^{(1)} | \boldsymbol{\theta}_{t-1}^{(2)})$.
- Update $\boldsymbol{\theta}_t^{(2)} \sim \pi(\boldsymbol{\theta}^{(2)} | \boldsymbol{\theta}_t^{(1)})$
- Return new state $\boldsymbol{\theta}_t = (\boldsymbol{\theta}_t^{(1)}, \boldsymbol{\theta}_t^{(2)})$

Iterating this process yields a Gibbs sampler.

More blocks $\rightarrow$ more conditional distributions.

Use and check output in the same way as for other MCMC methods.

Example: mean zero Bivariate Normal with covariance matrix ($-1 < \rho < 1$)

$$S = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

and log-target density kernel

$$\log g(\boldsymbol{\theta}) = -\frac{1}{2}\boldsymbol{\theta}^T S^{-1}\boldsymbol{\theta}$$

Blocks: (only one blocking possible) $\theta^{(1)} = \theta_1$, $\theta^{(2)} = \theta_2$

Conditionals:

$$\theta^{(1)}|\theta^{(2)} \sim N(\rho\theta^{(2)}, 1 - \rho^2)$$

$$\theta^{(2)}|\theta^{(1)} \sim N(\rho\theta^{(1)}, 1 - \rho^2)$$

The Gibbs cycles preserve the joint target distribution!

See R-code `mcmc_3.R`

OK, so the Gibbs cycles does not break joint target distribution; so what?

It is often possible to find a blocking so that all conditionals have a simple form! (i.e. allows easy characterization/sampling).

Example on previous slide: Goes from multivariate distribution (requires linear algebra etc) to two univariate normals which are trivial to sample.

In Bayesian statistics, one very often encounter/design model for similar simplifications, referred to as conditional conjugacy (will look at this later). Focus mainly of Gibbs for Bayes here...

Designing a good Gibbs sampler (an art form) requires the ability to recognize the conditionals from the original target distribution

Finding the conditionals

Suppose you start off with some target distribution with density $\pi(\boldsymbol{\theta}) = \pi(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})$ (and density kernel $g(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})$). The two conditionals are

$$\pi(\boldsymbol{\theta}^{(1)}|\boldsymbol{\theta}^{(2)}) = \frac{\pi(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})}{\pi(\boldsymbol{\theta}^{(2)})} = \frac{g(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})}{C\pi(\boldsymbol{\theta}^{(2)})}$$

$$\pi(\boldsymbol{\theta}^{(2)}|\boldsymbol{\theta}^{(1)}) = \frac{g(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})}{C\pi(\boldsymbol{\theta}^{(1)})}$$

Notice that the denominators here do not depend on variable to be updated, only the variable we condition on.

We may write this as

$$\pi(\boldsymbol{\theta}^{(1)}|\boldsymbol{\theta}^{(2)}) \propto g(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})$$

Read this as: shape of $\pi(\boldsymbol{\theta}^{(1)}|\boldsymbol{\theta}^{(2)})$ (in $\boldsymbol{\theta}^{(1)}$) is the same as shape of $g(\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})$ (with $\boldsymbol{\theta}^{(2)}$ kept fixed).

Completely same reasoning for $\pi(\boldsymbol{\theta}^{(2)}|\boldsymbol{\theta}^{(1)})$.

Example above

Background: (no need to find a)

$$\log p(x) = a + bx + cx^2 \Rightarrow x \sim N\left(-\frac{b}{2c}, -\frac{1}{2c}\right)$$

To find the $\theta^{(1)}|\theta^{(2)}$, start with log-density kernel:

$$\log g(\theta_1, \theta_2) = -\frac{\theta_1^2 + \theta_2^2 - 2\rho\theta_1\theta_2}{2(1-\rho^2)}$$

Considering the shape of the above expression *as a function of* θ_1 *only*, we obtain that

$$\log \pi(\theta_1|\theta_2) = \text{constant} - \frac{\theta_1^2 - 2\rho\theta_1\theta_2}{2(1-\rho^2)}$$

$$= \text{constant} + b\theta_1 + c\theta_1^2,$$

where

$$b = \frac{2\rho\theta_2}{2(1-\rho^2)},\ c = -\frac{1}{2(1-\rho^2)}$$

$$\Rightarrow \pi(\theta_1|\theta_2) = \mathcal{N}(\theta_1|\rho\theta_2, 1-\rho^2)$$

Notice, no need to find the normalizing constant!

More rules for recognizing distributions

Normal is the exponential of a quadratic polynomial

$$\log p(x) = a + bx + cx^2 \Rightarrow x \sim N\left(-\frac{b}{2c}, -\frac{1}{2c}\right)$$

Exponential distribution is the exponential of linear polynomial

$$\log p(x) = a + bx,\ x > 0\ \Rightarrow x \sim Exp(-1/b)$$

Gamma:

$$\log p(x) = a + b\log(x) + cx,\ x > 0$$

$$\Rightarrow x \sim Gamma(b+1, -1/c)$$

Beta:

$$\log p(x) = a + b\log(x) + c\log(1-x)\ 0 < x < 1$$

$$\Rightarrow x \sim Beta(b+1, c+1)$$

Inverse Gamma (wikipedia):

$$\log p(x) = a + b \log(x) - c/x,\ x > 0$$

$$x \sim InverseGamma(-(b+1), c)$$

Multivariate Normal (**b**; vector **c** symmetric positive definite matrix)

$$\log p(\mathbf{x}) = a + \mathbf{b}^T \mathbf{x} + \frac{1}{2} \mathbf{x}^T \mathbf{c} \mathbf{x}$$

$$\Rightarrow x \sim N(-\mathbf{c}^{-1}\mathbf{b}, -\mathbf{c}^{-1})$$

Further example: Normal observations, mean, precsion

Data: $y_i|\mu,\tau \sim N(\mu,\tau^{-1}),\ i = 1,\dots,n$

Prior, $\mu \sim N(0, 10^2)$

Prior, $\tau \sim Gamma(1, 1) = Exp(1)$

Collection of parameters $\boldsymbol{\theta} = (\mu, \tau)$

Wish to simulate from the posterior $\pi(\mu, \tau|\mathbf{y})$ using Gibbs sampling.

Recall

$$g(\boldsymbol{\theta}) = L(\boldsymbol{\theta})p(\boldsymbol{\theta})$$

Start with the likelihood

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{\tau}{2}(y_i - \mu)^2\right)$$

$$\propto \tau^{n/2} \exp\left(-\frac{\tau}{2}\sum_{i=1}^{n}(y_i - \mu)^2\right)$$

Take log

$$\log L(\boldsymbol{\theta}) = \text{const} + \frac{n}{2}\log(\tau) - \frac{\tau}{2}\sum_{i=1}^{n}(y_i - \mu)^2$$

Then the priors

$$\log p(\boldsymbol{\theta}) = \text{const} - \frac{\mu^2}{2 \cdot 100} - \tau,\ \tau > 0$$

Hence (omitting normalizing constants)

$$\log g(\mu, \tau) = \frac{n}{2}\log(\tau) - \frac{\tau}{2}\sum_{i=1}^{n}(y_i - \mu)^2 - \frac{\mu^2}{2 \cdot 100} - \tau$$

For the Gibbs sampler, we need the conditional posteriors $\mu|\tau, \mathbf{y}$ and $\tau|\mu, \mathbf{y}$:
First $\mu|\tau, \mathbf{y}$ (omit everything not depending on μ):

$$\log \pi(\mu|\tau, \mathbf{y}) = \text{const} - \frac{\tau}{2}\sum_{i=1}^{n}(y_i - \mu)^2 - \frac{\mu^2}{2 \cdot 100}$$

Log-density a quadratic function in $\mu \Rightarrow$ must be normal!
Trivial calculations (finding b and c yields)

$$\mu|\tau, \mathbf{y} \sim N\left(\frac{\tau \sum_{i=1}^{n} y_i}{n\tau + 1/100}, \frac{1}{n\tau + 1/100}\right)$$

Second, find distribution of $\tau|\mu, \mathbf{y}$ (omit everything not depending on τ)

$$\log \pi(\tau|\mu, \mathbf{y}) = \text{const} + \frac{n}{2}\log(\tau) - \frac{\tau}{2}S^2 - \tau$$

where $S^2 = \sum_{i=1}^{n}(y_i - \mu)^2$.
Go back to list above: Shape of log-conditional posterior is $a + b\log(\tau) + c\tau$: must be a Gamma distribution!

$$b = n/2,\ c = -S^2/2 - 1$$

$$\Rightarrow \tau|\mu, \mathbf{y} \sim Gamma(n/2 + 1, 1/(S^2/2 + 1))$$

Notice, big complicated joint posterior reduces to two conditional posteriors which both are straight forward to sample from!

See R-code `mcmc_3.R`

Conditional conjugacy

The example in the previous slides illustrates the concept of conditional conjugacy, namely that *the distribution of the prior and the conditional posterior are from the same family* (Normal for μ and Gamma for τ).

Since we may often choose the family of the prior-distribution rather freely, we often choose the prior to be conjugate with the likelihood under consideration, as this simplifies the Gibbs sampler.

See a more comprehensive list of conjugate distributions at Wikipedia.

Metropolized Gibbs sampler

Still it is not always possible to find priors that give closed form distributions for the conditional posteriors. In this case, it is still allowed to perform MH steps which targets the conditional posterior, in stead of exact updates. This tends to deteriorate the performance of the sampler somewhat, and also introduces the need to tune the MH step.

Typically, a "real life" Gibbs sampler consist of the majority of blocks being updated using closed form distributions (normal, Gamma etc.) and a few, low-dimensional blocks using MH steps.

Worked example: AR(1) model

$$y_1|\tau,\phi \sim N(0, (\tau(1-\phi^2))^{-1})$$

$$y_t|y_{t-1},\tau,\phi \sim N(\phi y_{t-1}, \tau^{-1}),\ t = 2,\ldots,T$$

$$\tau \sim Gamma(1,1),\ \phi \sim Uniform(-1,1)$$

$$L(\boldsymbol{\theta}) = \sqrt{\frac{\tau(1-\phi^2)}{2\pi}} \exp\left(-\frac{y_1^2\tau(1-\phi^2)}{2}\right)$$

$$\times \prod_{t=2}^{T} \sqrt{\frac{\tau}{2\pi}} \exp\left(-\frac{(y_t-\phi y_{t-1})^2\tau}{2}\right)$$

τ-conditional posterior:

$$\tau|\phi,\mathbf{y} \sim Gamma\left(T/2+1, 1/d\right)$$

$$d = 1 + \frac{y_1^2(1-\phi^2)}{2} + \frac{1}{2}\sum_{t=2}^{T}(y_t-\phi y_{t-1})^2$$

I.e. conjugate prior, easy updating.

BUT:

$$\log \pi(\phi|\tau, \mathbf{y}) = \text{const} + \frac{1}{2}\log(1-\phi^2) - \frac{y_1^2 \tau (1-\phi^2)}{2}$$

$$-\frac{\tau}{2}\sum_{t=2}^{T}(y_t - \phi y_{t-1})^2, \ -1 < \phi < 1.$$

"Almost quadratic" in ϕ, but the term $\frac{1}{2}\log(1-\phi^2)$ is not quadratic!

Idea; use the shape of $-\frac{\tau}{2}\sum_{t=2}^{T}(y_t - \phi y_{t-1})^2$ as the proposal of an independent MH step. This term will the most important unless $|\phi|$ is very close to 1.

Rewrite the term:

$$-\frac{\tau}{2}\sum_{t=2}^{T}(y_t-\phi y_{t-1})^2 = a+\phi \underbrace{\tau\sum_{t=2}^{T} y_t y_{t-1}}_{=b} -\phi^2 \underbrace{\frac{\tau}{2}\sum_{t=1}^{T-1} y_t^2}_{=-c}.$$

Suggests a

$$N(-\frac{b}{2c},-\frac{1}{2c}) = N\left(\frac{\sum_{t=2}^{T} y_t y_{t-1}}{\sum_{t=1}^{T-1} y_t^2}, \frac{1}{\tau\sum_{t=1}^{T-1} y_t^2}\right)$$

proposal in the independent MH step.

See R-code `mcmc_3.R`

Predict future outcome 100 days ahead

See R-code `mcmc_3.R`

Directions for implementing efficient Gibbs samplers

make blocks as large as possible E.g. if it is possible to update a complete multivariate normal, do so rather than many univariate normal updates.

make blocks as independent as possible. Strong dependencies of variables in different blocks tends to give poor Gibbs sampler performance. I.e. try to keep strongly dependent variables in the same block.

Spend time obtaining efficient Metropolized updates of as low dimension as possible.